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Modeling and finite element simulation of polypropylene behavior under severe plastic deformation by high pressure torsion

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Abstract — SPD technology, also known as NanoSPD, is the most efficient way of producing nanomaterials. However, two processing methods have been investigated more intensively: high-pressure torsion (HPT) and equal channel angular pressing (ECAP). Today, the high pressure torsion is one of the most promising methods for producing large shear strain, and has been the subject of widely investigations as a new processing method for nanostructured materials due to their capability to apply infinitely large strains to the material and to deform materials to very high plastic deformations without significantly altering the cross-section of the sample. In this paper, a phenomenological elastic-viscoplastic constitutive model was experimentally identified, and, coupled with the three-dimensional finite element method (FEM) to examine the different processing parameters governing the deformation behavior of polypropylene (PP) during HPT process. The plastic deformation behavior of polypropylene (PP) resulting from HPT process was investigated. To this end, the impact of various HPT process parameters, such as phases enchainment or choice of sequence, imposed compressive displacement (pressure), torsion angle, angular velocity angle and sample dimensions, is presented. Based on the numerical results obtained, recommendations on process conditions were formulated at the end of this work.

Keywords: *Severe plastic deformation, High pressure torsion, Polypropylene, Finite element method, Constitutive modeling.*

Nomenclature

Symbol	Designation
b, c, R_1, γ	Hardening parameters
$\mathbf{C}$	Fourth-order isotropic elastic modulus tensor
$\mathbf{d}$	Strain rate tensor
d	Distance along sample thickness from the bottom
$\mathbf{d}^e$	Elastic part of strain rate tensor
$\mathbf{d}^p$	Viscoplastic part of strain rate tensor
dl	Arc element of the disc
E	Young's modulus
h	Thickness of the disc
$H(w)$	Heaviside function
$\mathbf{I}$	Unity tensor
k	Initial yield stress in Chaboche model
K	Viscoplastic resistance
n	Rate sensitivity coefficient
N	Number of turns
$\dot{p}$	Equivalent viscoplastic strain rate
r	Radius of the disc
R	Drag stress
$\mathbf{W}$	Spin tensor
$\mathbf{X}'$	Kinematic hardening tensor $\mathbf{X}$
δ_{ij}	Kronecker symbol
$\{\varepsilon\}$	Strain vector
ε	Strain
ε_0	Initial yield strain
ε^p	Plastic strain
$\dot{\varepsilon}$	Strain rate
ε_{eq}	Equivalent plastic strain
ε_{true}	Logarithmic true strain
λ, μ	Lamé constants
γ_{total}	Total shear strain
θ	Rotation angle
$\tilde{\sigma}$	Jaumann derivative of the Cauchy stress tensor

σ'	Deviatoric part of the stress tensor σ
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I. Introduction

In recent years, the severe plastic deformation (SPD) technology is one of the most successful procedures to produce ultrafine-grained (UFG) and nanostructured materials [1–3]. The SPD processes have been extensively studied over the last three decades to enable the manufacturing of UFG materials with unique properties [4,5]. Nowadays, the SPD methods are used not only for the modification of existing materials, but they are also used to design and synthesize new materials with advanced mechanical and functional properties [6]. The SPD methods were used to synthesize new materials or to stabilize metastable phases SPD methods such as Equal Channel Angular Pressing [7], High Pressure Torsion [8], Twist Extrusion [9], Accumulative Spin Bonding [10], Accumulative back extrusion [11], Cyclic Rotating Bending [12] and Nonlinear Rotary Extrusion [13] are still the most popular techniques to produce UFG metallic materials. In addition to these techniques, there are now over 60 SPD methods or their modifications, including Repetitive Upset Extrusion [14], Friction Stir Processing (FSP) [15], Cyclic Extrusion and Compression [16], Hydrostatic Extrusion [17], High Pressure Sliding [18], High-Pressure Double Torsion [19], High Pressure Tube Twisting [20], Torsional-equal channel angular pressing [21], etc. [22,23].

In the early studies of UFG microstructure formation during SPD, two processing methods have been investigated more intensively which are Equal Channel Angular Pressing (ECAP) and High Pressure Torsion (HPT) [24]. Moreover, these methods have been applied to a wide range of materials such as metals, composites, ceramics, glasses, polymers, functionally graded materials [25]. As discussed above, the ECAP and HPT methods have made a large contribution to the development of the NanoSPD field [26–28]. Compared to ECAP and other Classic SPD methods, HPT is especially effective to introduce extremely large shear strain [29–33].

Indeed, the High Pressure Torsion (HPT) is a widely used process for modifying the materials structure and properties. It was introduced by Bridgman in 1935 [34]. He was the first scientist who could achieve in laboratory high pressure torsion (HPT) [35]. The principle of HPT is a severe plastic deformation process which a high pressure with torsion is applied simultaneously. The anvils are provided with cylindrical grooves and the sample (usually in the form of disc) is placed between the anvils. The disc is torsionally strained under high pressure. Furthermore, the large shear strain is induced by rotating the anvils with respect to each other (Fig. 1). In general, there are two distinct types of HPT processing depending on the shape of the anvils: constrained and unconstrained HPT [36]. In constrained HPT, a specimen is placed into the cavity of the lower anvil or both anvils [37], which can prevent material to flow outward. However, in unconstrained

HPT, a specimen is placed between two anvils of HPT tool and subjected to high pressure [38], in which, the sample material flow freely when the high pressure is applied.

In recent years, many authors have shown an increased interest in SPD processing of polymers, and a few amounts of literature has been published on the HPT of polymers [39,40]. It was shown that monomers or monomer mixtures polymerize when deformed by high pressure torsion [41,42]. Therefore, the HPT process was shown to cause a considerable change in the polymer morphology both at the micro and molecular scale [43,44]. Moreover, these changes directly influence the mechanical properties of polymers such as strength, elastic modulus, fracture toughness, etc. [45-48].

It is important to note that the FEM simulations have been shows that the plastic strains distribution is inhomogeneous over the sample thickness [40,49,50]. It should be noted that the HPT technique is especially attractive due to the possibilities of better controlled processing parameters such as, applied pressure, torsion angle, temperature, strain rate, etc. [40,51-65]. In this context, to illustrate the plastic flow and the distribution of stresses and strains, the finite element method (FEM) has been used in the present work to understand the materials behavior during HPT process.

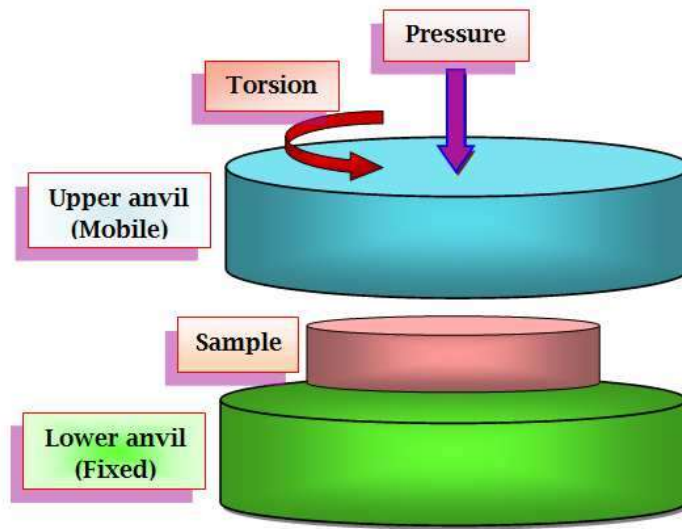


Fig. 1 Schematic geometry of HPT process

II. Analytical determination of equivalent plastic strain

In HPT process, as shown in figure 2, the shear strain is calculated using following equation [66]:

$$d\gamma_{total} = \frac{dl}{h} = \frac{rd\theta}{h} \quad (1)$$

where γ_{total} is the total shear strain applied, r and h are the radius and thickness of the disc, respectively.

By assuming that the thickness of the disc h is independent of the rotation angle θ , it follows from formal integration of equation (1) that since $\theta = 2\pi N$, the total shear strain γ_{total} can be expressed as:

$$\gamma_{total} = \frac{2\pi N.r}{h} \quad (2)$$

where N is the number of revolutions. The equivalent plastic strain is calculated using a commonly used relationship:

$$\varepsilon_{eq} = \frac{\gamma_{total}}{\sqrt{3}} = \frac{2\pi N.r}{\sqrt{3}.h} \quad (3)$$

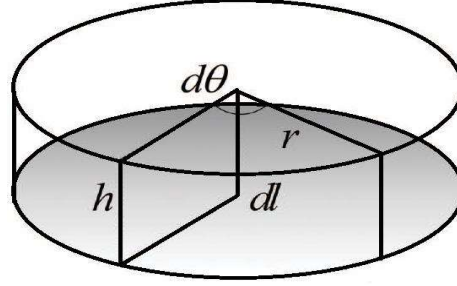


Fig. 2 A schematic view of the sample dimensions and parameters used to calculate the equivalent strain in HPT process

We can obtain the logarithmic true strain, ε_{true} , as follows:

$$\varepsilon_{true} = \ln(1 + \varepsilon) = \ln\left(1 + \frac{2\pi N.r}{\sqrt{3}.h}\right) \quad (4)$$

Practically, the equation (4) can be simplified as [67]:

$$\varepsilon_{true} = \ln\left(\frac{2\pi N.r}{h}\right) \quad (5)$$

As a result, the strain during HPT is equal to zero at the center of the sample and increases linearly until reaching a maximum near the edges. Thus, the microstructure produced by high pressure torsion is heterogeneous. Some authors reported that the microhardness also varies significantly along the disc radius processed by HPT [68-70].

III. Constitutive equations of the used model

To study the behaviour of polypropylene during HPT process, a phenomenological model of Chaboche is identified. It should be noted that the large strain behaviour of the studied material is characterized by strain-rate dependent yield followed by a strain hardening. Then, to account for the effect of the above parameters, kinematic hardening is added to the model [71,72]. The strain rate tensor $\mathbf{d}$ is decomposed into an elastic part $\mathbf{d}^e$ and a viscoplastic part $\mathbf{d}^{vp}$ as:

$$\mathbf{d} = \mathbf{d}^e + \mathbf{d}^{vp} \quad (6)$$

The elastic strain rate tensor $\mathbf{d}^e$ is given by the hypoelastic law:

$$\mathbf{d}^e = \mathbf{C}^{-1} \tilde{\boldsymbol{\sigma}} \quad (7)$$

where $\tilde{\boldsymbol{\sigma}} = \dot{\boldsymbol{\sigma}} - \mathbf{W}\boldsymbol{\sigma} + \boldsymbol{\sigma}\mathbf{W}$ is the Jaumann derivative of the Cauchy stress tensor $\boldsymbol{\sigma}$ based upon the spin tensor $\mathbf{W}$, and

$\mathbf{C}$ is the fourth-order isotropic elastic modulus tensor:

$$C_{ijkl} = \frac{E}{2(1+\nu)} \left[(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) + \frac{2\nu}{1-2\nu} \delta_{ij}\delta_{kl} \right] \quad (8)$$

with E and ν are the Young's modulus and Poisson's ratio, respectively and δ is the Kronecker delta symbol. The

viscoplastic strain rate tensor $\mathbf{d}^{vp}$ can be expressed as:

$$\mathbf{d}^{vp} = \frac{3}{2} \dot{p} \frac{\boldsymbol{\sigma}' - \mathbf{X}'}{J(\boldsymbol{\sigma} - \mathbf{X})} \quad (9)$$

where $J(\boldsymbol{\sigma} - \mathbf{X})$ is defined by von Mises criterion:

$$J(\boldsymbol{\sigma} - \mathbf{X}) = \left[\frac{3}{2} (\boldsymbol{\sigma}' - \mathbf{X}') : (\boldsymbol{\sigma}' - \mathbf{X}') \right]^{\frac{1}{2}} \quad (10)$$

with $\boldsymbol{\sigma}' = \boldsymbol{\sigma} - \text{tr}(\boldsymbol{\sigma})/3\mathbf{I}$ and $\mathbf{X}'$ are respectively the deviatoric part of the stress tensor $\boldsymbol{\sigma}$ and the kinematic hardening tensor $\mathbf{X}$, and $\dot{p}$ is the equivalent viscoplastic strain rate written as:

$$\dot{p} = \left\langle \frac{J(\boldsymbol{\sigma} - \mathbf{X}) - R - k}{K} \right\rangle^n \quad (11)$$

The brackets are defined by $\langle w \rangle = wH(w)$. $H(w)$ is the Heaviside function ($H(w) = 0$ if $w < 0$, $H(w) = 1$ if $w \geq 0$). k is the initial yield stress, K is the viscoplastic resistance, n is the rate sensitivity coefficient and R is the isotropic internal stress or the drag stress. The hardening is controlled by $\dot{\mathbf{X}}$ and $\dot{R}$ which are given by:

$$\dot{\mathbf{X}} = \frac{C}{R+k} (\boldsymbol{\sigma} - \mathbf{X}) \dot{p} - \gamma \mathbf{X} \dot{p} \quad (12)$$

$$\dot{R} = b(R_1 - R) \dot{p} \quad (13)$$

where b , R_1 , c and γ are hardening parameters.

IV. Identification of the model parameters

To identify the model parameters, the experiments were conducted using cylindrical samples in polypropylene (PP) (20 mm in length and 10 mm in thickness) provided by the Goodfellow Company (United Kingdom), with a weight-average molar weight of 180 kg/mol. Its thermal properties were obtained by differential scanning calorimetry (DSC) analysis. Furthermore, the glass transition temperature of PP is about -20°C and the melting temperature is about 170°C , the amorphous phase at a room temperature is therefore in the rubbery state. The mechanical properties of polypropylene (PP) are summarized in table 1.

Table 1 Mechanical properties of polypropylene (PP)

Properties	Values
Density (g.cm^{-3})	0.9
Friction coefficient	0.1-0.3
Rockwell hardness	80-100
Shock resistance (Izod) (J m^{-1})	20-100
Poisson's ratio	0.40
Tensile modulus (GPa)	0.9-1.5
Tensile strength (MPa)	25-40

To identify the material parameters of the Chaboche model, compression tests at different strain rates (10^{-3} , 10^{-4} and 10^{-5} s^{-1}) were carried out on polypropylene specimens at room temperature using an Instron[®] electromechanical testing machine. The obtained stress-strain curves are shown in Figure 3. The material parameters of the elastic-viscoplastic constitutive model were determined from a least-squares regression fitting of the compression tests data. The values of the identified parameters are shown in table 2 for the studied materials. Figure 3 shows a good agreement between the constitutive model and experimental results.

Table 2 Identification of material parameters of polypropylene behaviour at room temperature

Parameters	Values of parameters
	1100 MPa
ν	0.4
k	10 MPa
σ_0	30.2 MPa
n	6.9
D	65
R_1	18 MPa
c	15 MPa
γ	-3.2

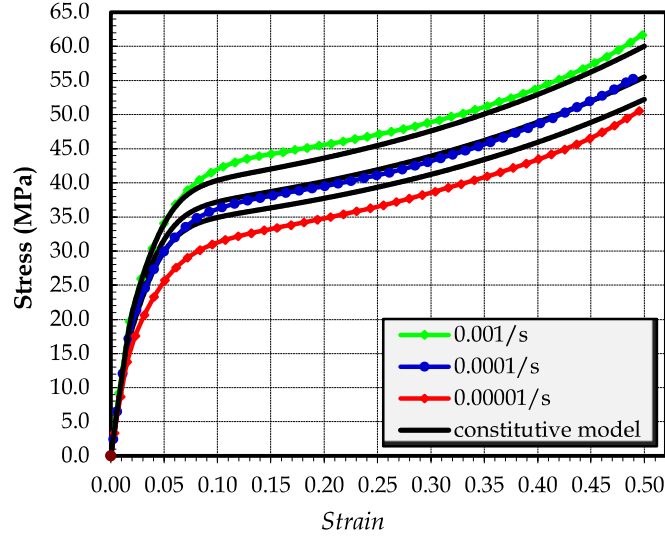


Fig. 3 Stress-strain curves obtained by compressive tests and constitutive model for polypropylene behaviour at room temperature

V. Three-dimensional finite element simulations of HPT process

The simulations were performed using the finite element software MSC.Marc. The initial dimensions of the polypropylene sample are 20 mm in diameter and 10 mm in thickness as shown in Fig.1. The upper and lower anvils were considered as rigid bodies in the modeling. In the first step, an imposed vertical displacement was assigned to the upper anvil in the compressive direction and was maintained during the second step of the rotation (torsion).

V.1. Meshes sensitivity effect

For reasons of symmetry in the loading and geometry of the sample, the quarter instead of the full sample, as illustrate in Figure 4, can be used. To study the mesh effect on the plastic deformation, five types of meshes of the quarter and the full disc were examined using 27-node hexahedral elements (see table 3). A vertical compressive displacement of 1mm, then a torsion angle of 60° were applied by the upper anvil on the HPT sample. Noting that the pressure was maintained during the rotation. To conduct the process in isothermal conditions and neglect the heat due to plastic deformation, the simulations were carried out at a weak angular velocity of 0.065 rad/s.

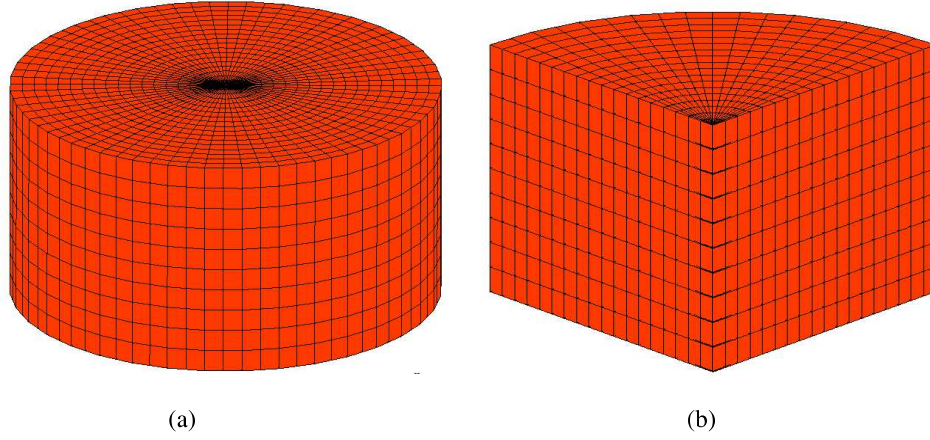


Fig. 4 Finite element discretization of: (a) a complete, and (b) a quarter of the disc

From a practical point of view, it is useful to confirm which of the meshes is efficient for obtaining reliable results. To this end, a comparison between the results obtained using both geometries and different meshes has been presented in table 3. Note that there is a slight difference between the results obtained by both geometries when the same size of the element is used for both geometries (see Tab. 3 and Fig. 5).

Table 3 Comparison of maximum plastic deformation values and CPU times for different full-disc and quarter-disc meshes

Mesh quality	Shape of sample	Number of elements	$\mathcal{E}_{max}$	CPU Time (sec)	Mesh quality
Mesh 1 (very coarse)	Complete disc	1944	0.8069738	9356	Complete disc
	Quarter disc	486	0.8007105	3241	Quarter disc
Mesh 2 (Coarse)	Complete disc	3240	0.8858734	11681	Complete disc
	Quarter disc	810	0.8646584	4597	Quarter disc
Mesh 3 (Medium)	Complete disc	7776	0.9495673	21587	Complete disc
	Quarter disc	1944	0.9275745	8137	Quarter disc
Mesh 4 (Fine)	Complete disc	12960	1.096282	32576	Complete disc
	Quarter disc	3240	1.085190	21463	Quarter disc
Mesh 5 (very fine)	Complete disc	15552	1.134624	53653	Complete disc
	Quarter disc	3888	1.115711	29365	Quarter disc

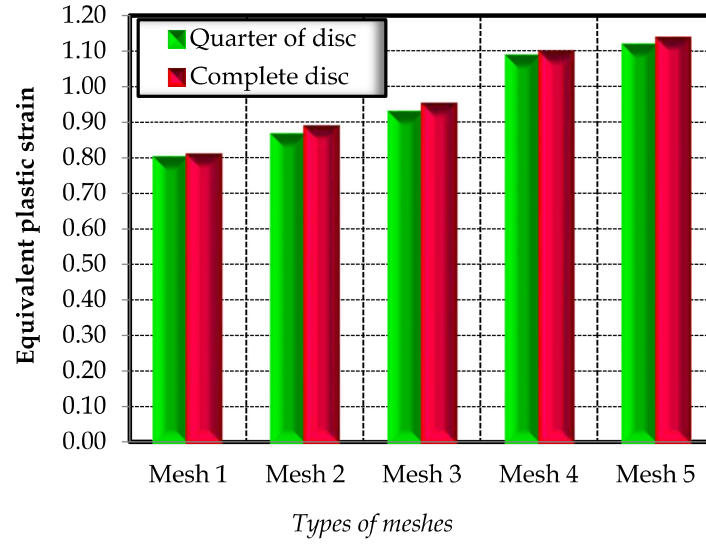


Fig. 5 Comparison of the maximal equivalent plastic strain distribution along the radial distance for both geometries (CD and QD) at the end of HPT process

Figure 6 illustrates the iso-value contours of the equivalent plastic strain obtained for the QD and CD discs at the end of HPT process. It can be observed that almost the same distribution of the equivalent plastic strain has been observed for both geometries.

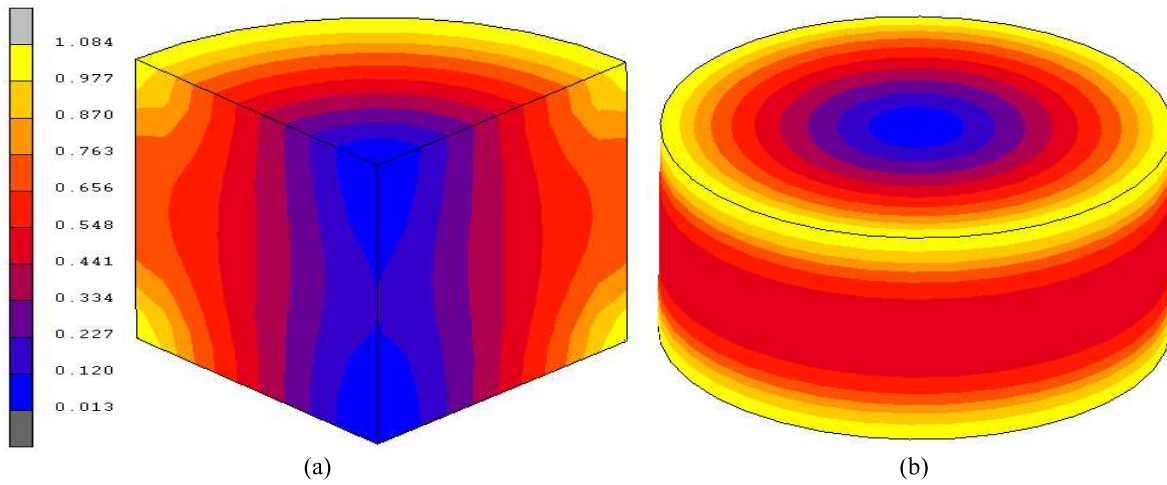


Fig. 6 Contour plot of equivalent plastic strain for (a) quarter of disc and (b) complete disc

It is important to note that the CPU time for the mesh 5 is about 1.5 times of that of mesh 4 in both cases (QD and CD).

Therefore, to obtain a best compromise between precision and computational time, in what follows, we use a complete disc with a fine mesh (**mesh 4**). This allows us to improve the exactitude without a significant increase in CPU time. It should be noted that from a practical point of view, it is useful to consider the entire sample to avoid any error which can be generated by symmetry conditions.

Figure 7 illustrates a comparison of the equivalent plastic strain distribution along the radial distance for the five meshes at the end of HPT process. It can be noted that the equivalent plastic strain increases with the increase of the number of elements especially at the periphery of the samples. In addition, a slight difference can be observed between the results obtained by mesh 4 (fine mesh) and mesh 5 (very fine mesh).

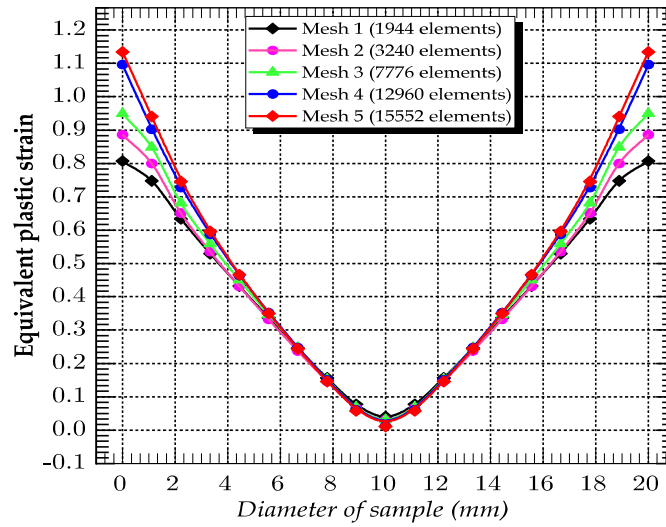


Fig. 7 Evolution of the equivalent plastic strain along the radial distance for the five meshes of the entire disc at the end of HPT process

V.2. Sequence effect

In most practical cases, we can distinguish two major scenarios or sequences to perform the HPT process. The first is to apply a compression followed by torsion using the upper anvil (**HPT process 1**). In the second, the sample is subjected simultaneously to compression-torsion (**HPT process 2**). To make a comparison between the both scenarios, a finite element simulation has been conducted using mesh 4 with 27-node hexahedral elements. The PP sample is subjected to compressive imposed displacement of 1 mm and torsion angle of 60°. The obtained results of the equivalent plastic strain distributions along the diameter and the thickness of PP sample are shown respectively in Figure 8, using both scenarios of HPT process. It can be easily observed that the plastic strain induced by the first scenario (**HPT process 1**) is higher

than that obtained by the second scenario (**HPT process 2**).

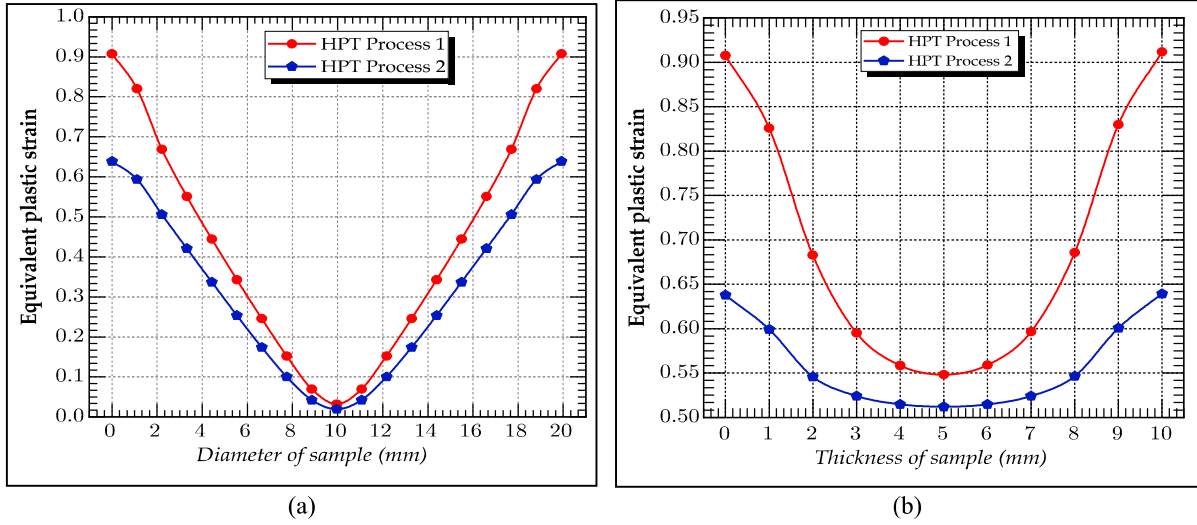


Fig. 8 Evolution of the equivalent plastic strain at the end of both scenarios of HPT process: (a) along the diameter and (b) the thickness of the disc

Furthermore, to give more information on the complete sample, the contour plots of the equivalent plastic strain at the end of HPT process in both cases are shown in Figure 9. It is obvious that the results of the equivalent plastic strain obtained using **HPT process 1** is greater than that obtained by **HPT process 2**. Consequently, in what follows the HPT process 1 is adopted to investigate the other parameters in this work.

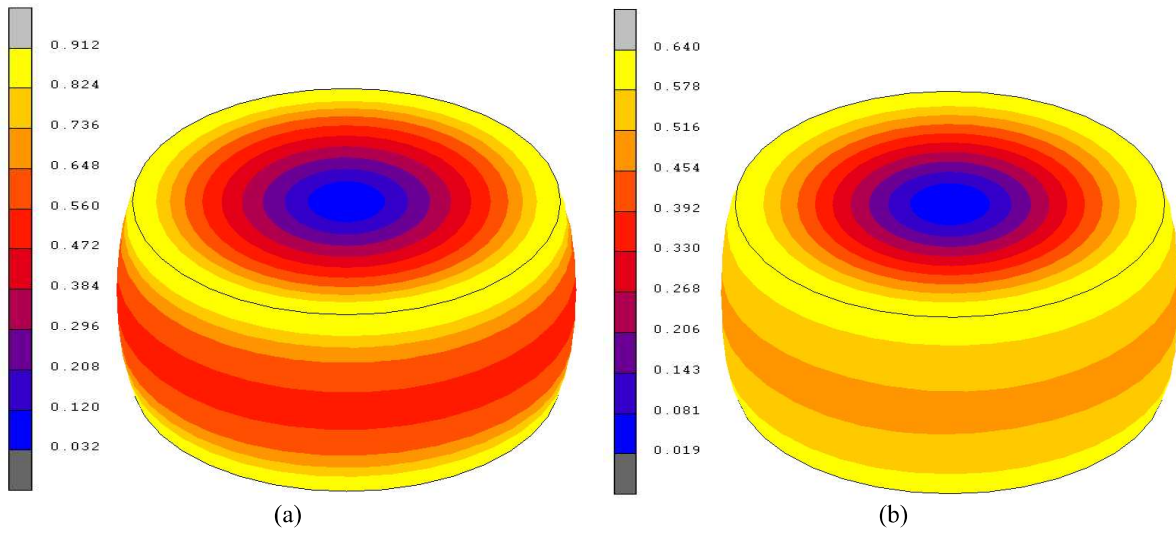


Fig. 9 Contour plot of the equivalent plastic strain in the cases of: (a) HPT process 1
and (b)- HPT process 2

V.3. Effect of the compressive imposed displacement

V.3.1. Evolution of the equivalent plastic strain

Because the mechanical properties of HPT materials are directly linked to the development of the plastic deformation induced into the sample, the analysis of this latter is presented here. The effect of the compressive imposed displacement on the evolution of the equivalent plastic strain along the radial distance is highlighted. To this end, the upper anvil of HPT process is subjected successively to five imposed vertical displacements of 0.4, 0.8, 1.2, 1.6 and 2.0 mm followed by a constant torsion angle of 30° with an angular velocity of 0.065 rad/s. The obtained results are shown in Figure 10. It is apparent that the equivalent plastic strain increases with the increase of the imposed compressive displacement. Indeed, for an imposed displacement of 0.4 mm, an equivalent plastic strain (ϵ_{eq}) of 0.338 was obtained, whereas, for the one of 2.0 mm, $\epsilon_{eq} \approx 1.195$, which is more than 3.5 times that of 0.4 mm.

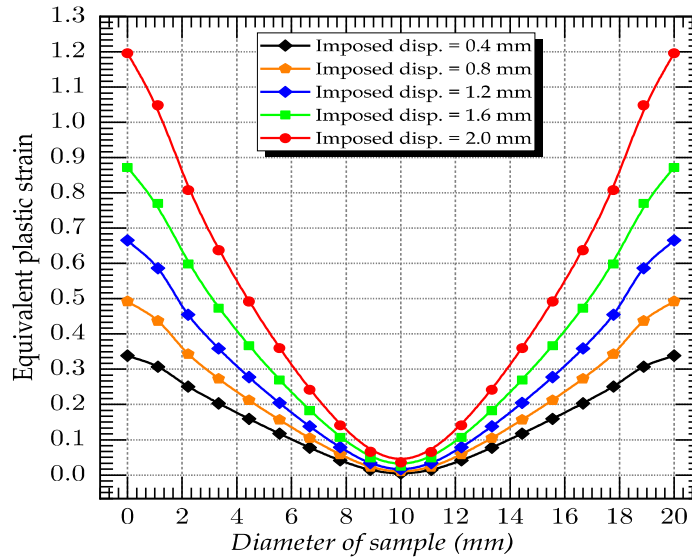


Fig. 10 Evolution of the equivalent plastic strain along the radial distance of the sample for different imposed vertical displacements

The contour plots of the equivalent plastic strain in PP sample at the end of HPT process for different imposed vertical displacements are shown in figure 11. It can be found that the influence of this parameter becomes significant as one moves away from the center of the sample, and that the same applies to the heterogeneity of the plastic strain distribution.

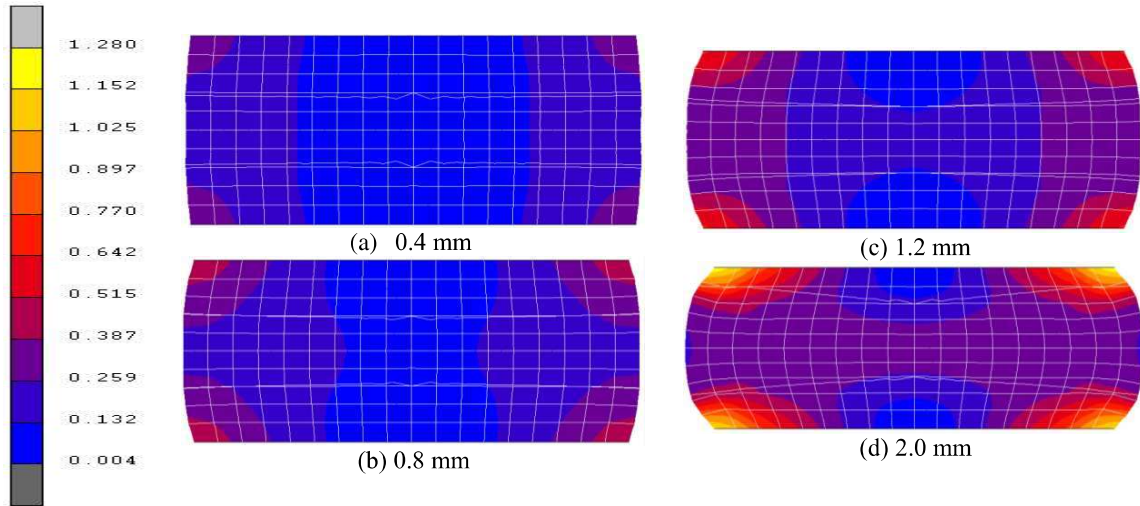


Fig. 11 Contour plot of the equivalent plastic strain in PP sample at the end of HPT process for different imposed vertical displacements: (a) 0.4 mm, (b) 0.8 mm (c) 1.2 mm and (d) 2.0 mm

V.3.2. Evolution of the von Mises stresses

In order to study the state of the stresses in polypropylene samples, the evolution of von Mises stresses along the diameter of the upper surface at the end of HPT process with different compressive imposed displacement are presented in figure 12. It can be observed that as the imposed displacement increases, the von Mises stresses increase, and become very important at the edge of the sample. Indeed, the maximum value of von Mises stress, 141.35 MPa, is obtained for the case of a compressive imposed displacement of 2 mm and is located at the peripheral region.

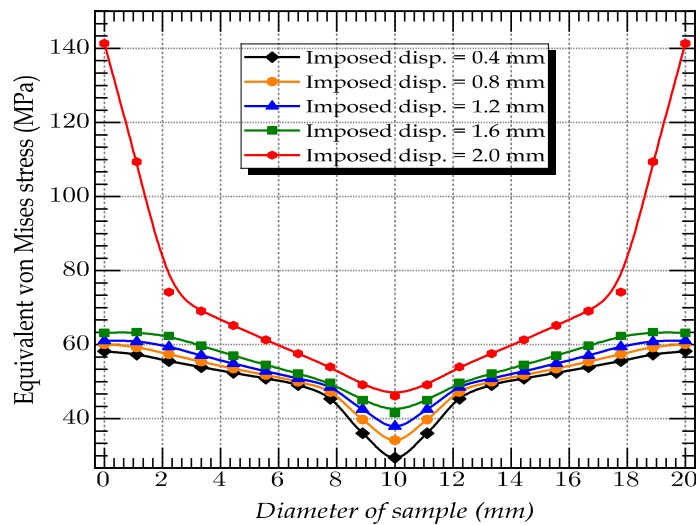


Fig.12 Evolution of von Mises stresses versus the radial distance on the upper face at the end of HPT process for different compressive imposed displacements

V.4. Effect of torsion angle

V.4.1. Evolution of the equivalent plastic strain

To optimize the HPT process, the effect of torsion angle is analyzed in this section. To this end, a vertical displacement of 1 mm was imposed by the upper anvil on the PP sample and was maintained during the torsion with different angles of (15°, 30°, 45°, 60°, 75° and 90°). The simulations were performed with an angular velocity of 0.065rad/s and compression speed of 0.5 mm/s. Figure 13 illustrates the evolution of the equivalent plastic strain along the radial distance of the sample for the different selected torsion angles. It is obvious that by increasing the rotation angle, the accumulated plastic strain is increased considerably. Furthermore, it becomes very important at the sample periphery, however at the middle region of the sample it remains always weak.

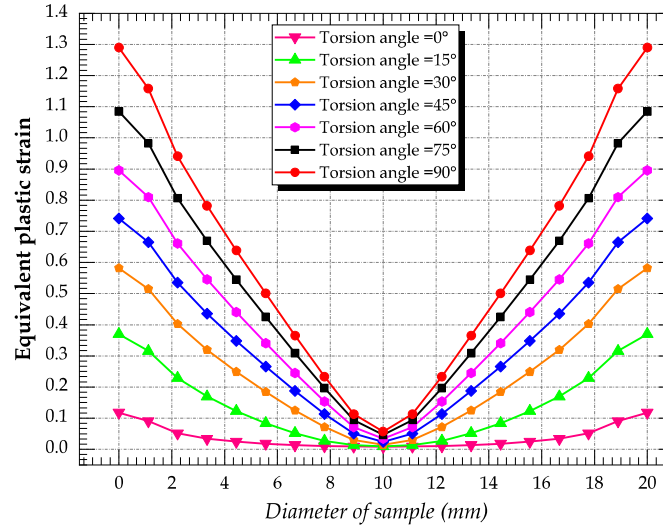


Fig.13 Evolution of the equivalent plastic strain versus the radial distance of the sample for different torsion angles (0°, 15°, 30°, 45°, 60°, 75° and 90°) combined with a compressive imposed displacement of 1 mm

Figure 14 shows the contour plots of the deformed geometry of polypropylene samples after HPT process for (a) a compressed state with an imposed displacement of 1.0mm without torsion and (b) an imposed displacement of 1mm with a torsion angle of 90°. It is worth to note, that after a simple compression only (Fig.14-a), the equivalent plastic strain remains very weak, and the center regions are less deformed than the edges of the sample in which the maximum equivalent plastic strain does not exceed 0.12. But, after applying a torsion angle of 90° with an imposed displacement compression of 1.0 mm as shown in figure 14-b, the equivalent plastic strain is relatively high for both regions and in

particular the region which are far from the center. Moreover, it may be concluded that in the case of PP samples, the radials non-uniform deformation is more pronounced than that of the axially non-uniform deformation.

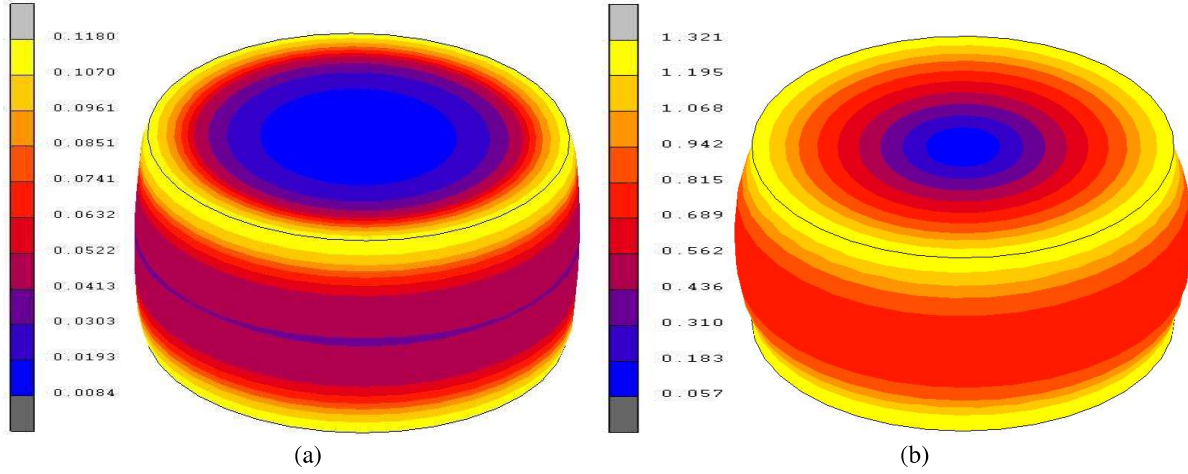


Fig. 14 Distribution of the equivalent plastic strain in the deformed samples after HPT process in the case of: (a)- 1 mm of compression without torsion and (b)- 1 mm of compression with a torsion angle of 90°

V.4.2. Evolution of the von Mises stress

Figure 15 depicts the estimation of von Mises stresses along the radial distance on the upper surface of the HPT sample using an imposed displacement of 1 mm with different torsion angle (15°, 30°, 45°, 60° and 90°). It has been shown, from this figure, that when the torsion angle increases, the von Mises stresses increase permanently. Likewise, when the radial distance from center increases, the von Mises stresses also increase considerably for different torsion angles. This can be justified that the material near the edge is processed under higher flow than that near the center region.

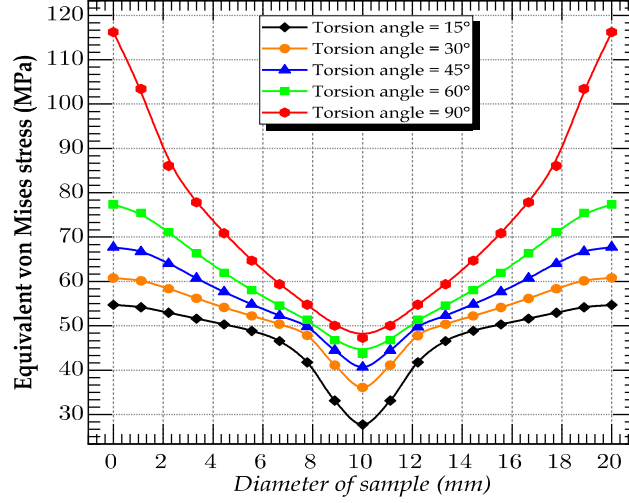


Fig. 15 Evolution of von Mises stresses along the radial distance on the upper surface at the end of HPT process for an imposed displacement of 1 mm with different torsion angles (15°, 30°, 45°, 60° and 90°)

V.4.3. Evolution of the equivalent plastic strain rate

It is important to know that the behavior of polymers strongly depends on the strain rate. Thus, to understand the heterogeneity in the plastic strain distribution in the polypropylene samples, a special attention has been given on the equivalent plastic strain rate distribution. Figure 16 illustrates the distribution of equivalent plastic strain rate at the end of HPT process of PP samples under a compressive imposed displacement of 1 mm with different torsion angles (15°, 30°, 45°, 60°, 75° and 90°). As seen, for all torsion angles, as the distance from the center increases, the plastic strain rate increases permanently. On other hand, it is apparent that the plastic strain rate increases considerably when the torsion angle becomes greater than 60°, but, when the torsion angle is less than 60°, the plastic strain rate increases slowly. In addition, it can be remarked that the maximum strain rate for the six torsion angles is located at the edge region. This aspect of heterogeneity can be attributed to a coupled effect of material behavior and the processing conditions, since the edges region of PP sample flow faster than the central portion.

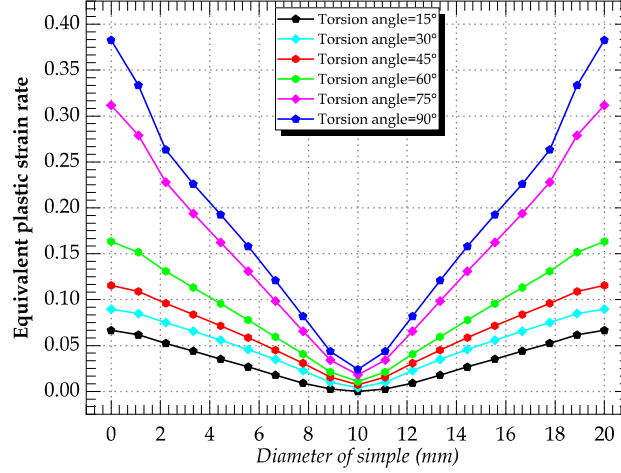


Fig. 16 Evolution of equivalent plastic strain rate along the radial distance on the upper surface of PP sample at the end of HPT process for an imposed displacement of 1 mm with different torsion angles (15°, 30°, 45°, 60°, 75° and 90°)

V.5. Combined effect of torsion angle and imposed displacement

Figure 17 exhibits the distribution of the equivalent plastic strain on the upper surface of the deformed sample during HPT process versus the time for different positions along the radius of the sample in the case of: (HPT-1): 2.0 mm of compression with a torsion angle of 30° and (HPT-2): 1 mm of compression with a torsion angle of 75°. It is apparent that the equivalent plastic strain increases away from the center of the sample for both cases. In addition, it can be observed that a low compressive displacement with a high torsion angle gives almost the same results as a high compressive displacement and a low torsion angle. Consequently, it can be concluded that by increasing the pressure and torsion angle, the plastic strain increases considerably.

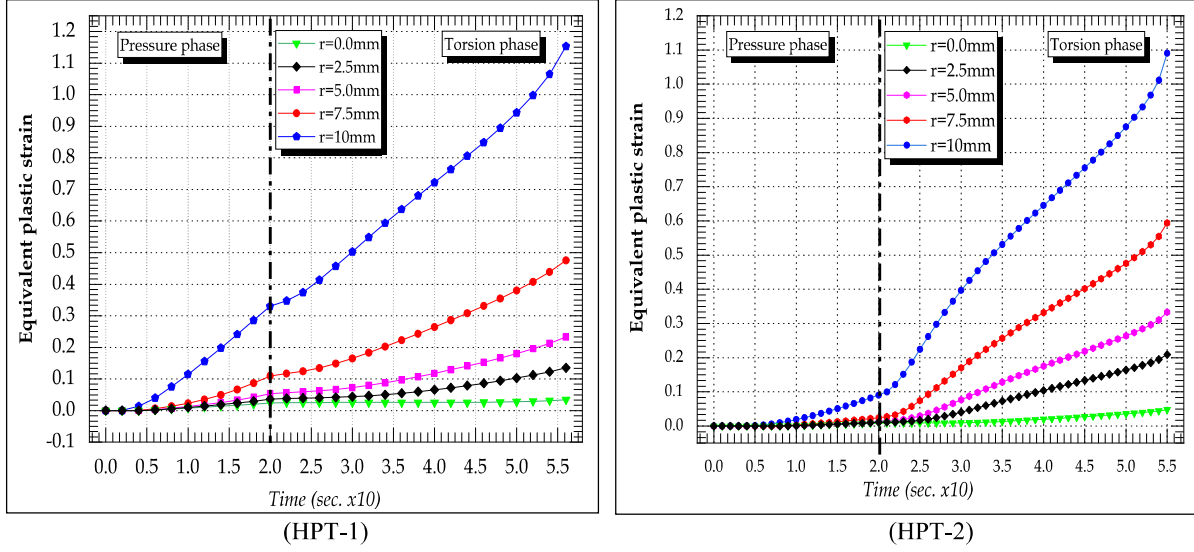


Fig. 17 Evolution of the equivalent plastic strain on the upper surface versus the time for different radial positions on the PP sample in the case of: (a) 2 mm of compression and a torsion angle of 30° , (b) 1 mm of compression and a torsion angle of 75°

V.6. Effect of the sample size (thickness-to-diameter ratio)

A considerable amount of literature has been published on the effect of the sample geometry (diameter and thickness). These studies have considered that this latter as a primordial parameter in experimentation and simulation of HPT process. Thus, to study the effect of sample geometry (diameter D and thickness t) on the plastic strain, five different ratios: $t/D = 0.2, 0.25, 0.3, 0.4$ and 0.5 have been considered. In these simulations, an imposed compressive displacement of 1 mm and an angular velocity of 0.065 rad/sec were used. Figure 18 shows the distribution of equivalent plastic strain on the upper face as a function of different ratios t/D at the end of HPT process. It is important to observe, as the thickness-to-diameter ratio increases, the equivalent plastic strain decreases.

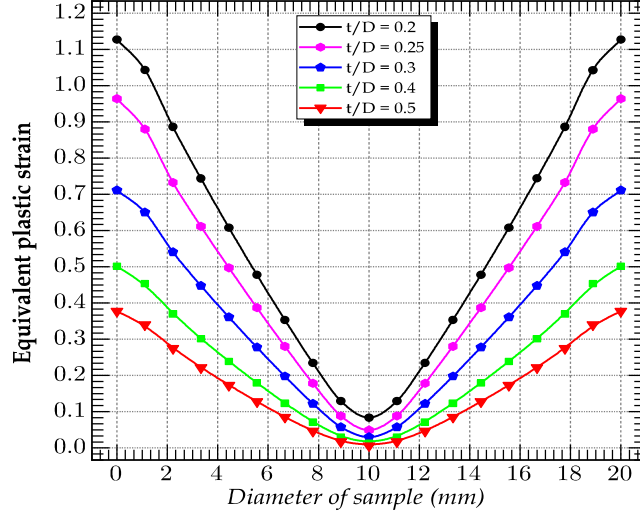


Fig. 18 Evolution of equivalent plastic strain along the radial distance of PP samples for different thickness-to-diameter ratios (t/D)

To illustrate this aspect, the contour plots of the equivalent plastic strain of undeformed polypropylene sample after HPT process for two different ratios: (a) $t/D=0.2$ and (b) $t/D=0.5$ have been presented in Figure 19. The large plastic deformation zone is concentrated near the edges of the disc and on the top and the bottom surfaces. Moreover, the results show that by increasing the sample radius, the equivalent plastic strain increases. As a result, it can be concluded from these simulations of HPT process, that to obtain a large plastic strain in the simple, it is advised to reduce the thickness-to-diameter ratio of HPT sample.

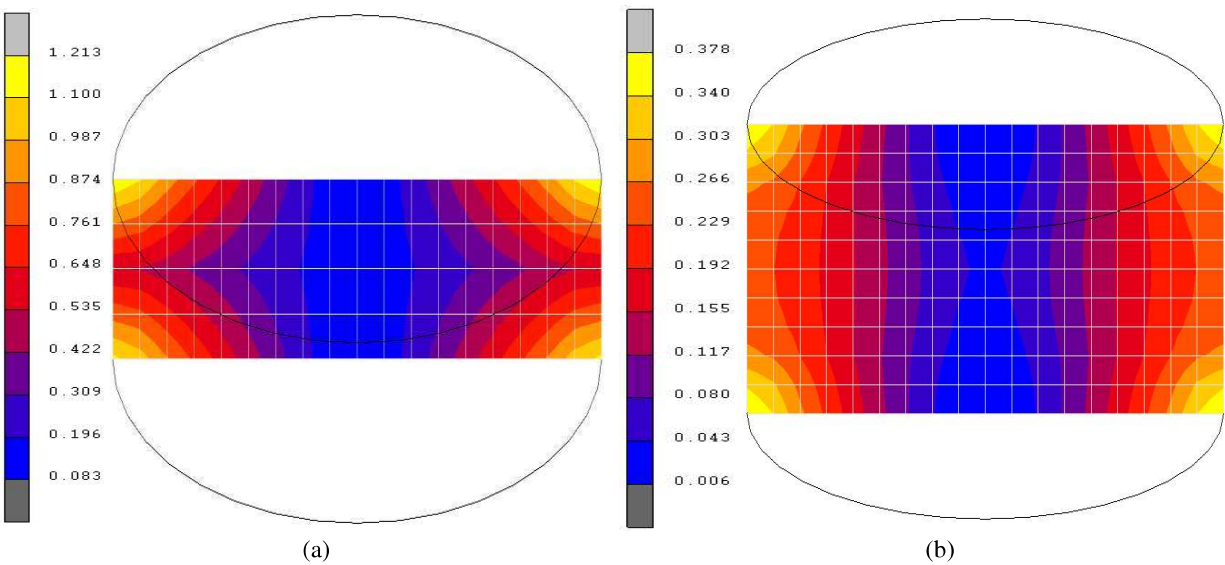


Fig. 19 Cutting planes plots of the equivalent plastic strain of PP samples along the thickness at the end of HPT process in the case of: (a) $t/D=0.2$, (b) $t/D=0.4$

V.7. Effect of the angular velocity

Because the behavior of polypropylene is very sensible to the rate deformation, the study of the effect of angular velocity on the polypropylene behavior during HPT process is addressed in this section. The sample is subjected to a constant compressive imposed displacement of 1 mm and torsion angle of 60° . Six angular velocities ($\pi/12$, $\pi/24$, $\pi/36$, $\pi/48$, $\pi/60$ and $\pi/72$ rad/s) corresponding respectively to torsion times of: 4, 8, 12, 16, 20 and 24 s are tested. The distribution of the equivalent plastic strain on the upper surface of PP samples at the end of HPT process versus the diameter for different angular velocities is plotted in figure 20. It can be noted that the results presented are almost the same for all angular velocities which means that the angular velocity has an almost negligible effect on the evolution of equivalent plastic strain. This can be attributed partially to the phenomenological model used, which does not consider the effect of temperature variation during plastic deformation (i.e., isothermal conditions).

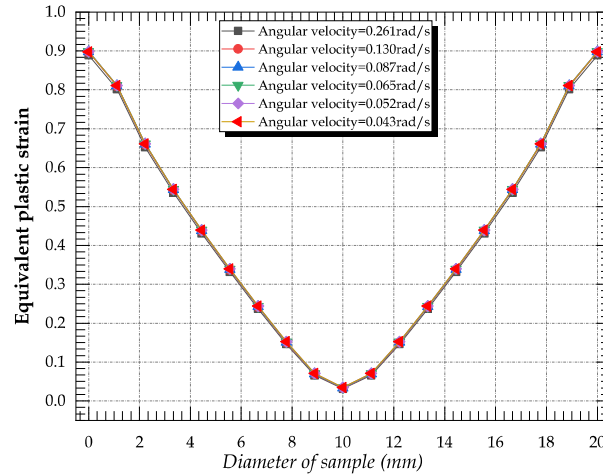


Fig. 24 Evolution of the equivalent plastic strain on the upper surface at the end of HPT process, versus the sample diameter for different angular velocities

VI. Conclusion

In this paper, the results of polypropylene behavior during severe plastic deformation by high pressure torsion process analysis using the three-dimensional finite element method are presented. The purpose of the present work is to study the

applicability of high pressure torsion process for a typical polymer which is polypropylene (PP). For this purpose, an identification of the parameters of Chaboche elasto-viscoplastic phenomenological model was performed using compression tests at different strain rates. After that, a numerical investigation was carried out to highlight the effects of the processing parameters on the polypropylene behavior as the scenario process, compressive imposed displacement, thickness-diameter ratio, torsion angle and angular velocity. To this end, the equivalent plastic strains and von Mises stresses distributions during HPT process have been presented and discussed.

Based on the obtained results, the following conclusions has been made:

- The first process (*HPT process 1*) which a compressive imposed displacement followed by torsion using the upper anvil is more effective than HPT process 2 where the compressive displacement and the torsion angle are applied simultaneously.
- A Hexahedral element class of Twenty-seven-node (HEX-27) and a fine mesh are recommended for this type of study, because they are computationally efficient to simulate the HPT process and to properly estimate the large plastic strain.
- It has been found that the equivalent plastic strain increases with the increase of the sample radius, the compressive imposed displacement, and the torsion angle.
- It should be noted that the equivalent plastic strain is lower at the center and increases by moving away from the center, which becomes very important at the periphery of the sample for all the studied parameters.
- The non-uniform of the equivalent plastic strain distribution in the sample is more pronounced in the radial direction compared to that of the axial direction.
- It is advised to use small thickness-diameter ratio to obtain a good homogeneity with a high level of plastic strain.
- It is found that by increasing the torsion angle, the strain rate increases considerably.
- The difference in the values of equivalent plastic strain between the center and the edge has been explained by the fact that the torsional strain is proportional to the distance from the center as given in Eq. (2), which means that the region of the sample periphery flows faster than that near the center.

Finally, it is important to point out that the plastic strain behavior of polypropylene during HPT is still not well understood, and that the effects of the process parameters such as number of rotations, strain rate, speed, sample thickness, temperature, etc. should be studied in more detail in relation to the aspect of microstructure orientation of HPT materials, and work in this direction is currently underway.

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Conflicts of Interest

The authors declare no conflict of interest. The founding sponsors had no role in the design of the study, in the collection, analyses or interpretation of data, in the writing of the manuscript and in the decision to publish the results.

References

1. Beygelzimer Y, Kulagin R, Estrin Y (2019) Severe Plastic Deformation as a Way to Produce Architected Materials. In *Architected Materials in Engineering and Nature*; Estrin, Y., Brechet, Y., Dunlop, J., Fratzl, P., Eds.; Springer: Berlin, Germany.
2. Durand-Charre M (2003) *The microstructure of steels and cast irons. History and interpretation.* Edition SIRPE; Paris (France); ISBN 2-906643-27-0.
3. Durand-Charre M (2014) *Damascus and pattern-welded steels, forging blades since the iron age.* Edition EDP Sciences; Les Ulis (France). Doi: 10.1051/978-2-7598-1635-4.
4. Bridgman PW (1952) *Studies in large plastic flow and fracture.* New York (NY): McGraw-Hill. Doi: 10.4159/harvard.9780674731349.
5. Sevillano JG, van Houtte P, Aernoudt E (1981) Large strain work hardening and textures. *Prog Mater Sci* 25:69-412. Doi: 10.1016/0079-6425(80)90001-8.
6. Valiev RZ, Kaibyshev OA, Kuznetsov RI, et al. (1988) Low temperature superplasticity of metallic materials. *Dokl Akad Nauk SSSR.* 301(4), 864-866.
7. Edalati K (2019) Metallurgical alchemy by ultra-severe plastic deformation via high-pressure torsion process. *Mater Trans* 60:1221–1229. Doi:10.2320/matertrans.MF201914.
8. Valiev RZ, Estrin Y, Horita Z, et al. (2006) Producing bulk ultrafine-grained materials by severe plastic deformation. *JOM* 58:33–39. Doi: 10.1007/s11837-006-0213-7.
9. Segal VM, Reznikov VI, Drobyshvskiy AE, et al. (1981) *Russ Metall.* 1:99-105. Doi: 10.1016/0921-5093(95)09705-8.
10. Bridgman PW (1935) Effects of high shearing stress combined Plastic working of metals by simple shear. with high hydrostatic pressure. *Phys Rev* 48:825-847. Doi: 10.1103/PhysRev.48.825.
11. Beygelzimer Y, Varyukhin V, Synkov S, et al. (1999) New techniques for accumulating large plastic deformations using hydroextrusion. *Phys Technol High Press* 9(3), 109-111.
12. Saito Y, Tsuji N, Utsunomiya H, et al. (1998) Ultra-fine grained bulk aluminum produced by accumulative roll-bonding (ARB) process. *Scr Mater* 39:1221–1227. Doi: 10.1016/S1359-6462(98)00302-9.
13. Valiahmetov OR, Galeyev RM, Salishchev GA (1990) Mechanical properties of VT 8 Ti-alloy of submicrocrystalline structure. *Fiz Met Metalloved* 10:204–206.

14. Huang JY, Zhu YT, Jiang HG, et al. (2001) Microstructures and dislocation configurations in nanostructured Cu processed by repetitive corrugation and straightening. *Acta Mater* 49:1497–1505. Doi: 10.1016/S1359-6454(01)00069-6
15. Kwon Y J, Shigematsu I, Saito N (2002) Production of ultra-fine grained aluminum alloy by friction stir process. *J Jpn Inst Met* 66:1325–1332. Doi: 10.2320/jinstmet1952.66.12_1325.
16. Richert J, Richert M, Zasadziński A, et al. (1979) A method of plastic processing of metals and alloys and a device for plastic processing of metals and alloys. PL Patent 123,026.
17. Kurzydłowski KJ (2006) Hydrostatic extrusion as a method of grain refinement in metallic materials. *Mater Sci Forum* 503-504:341–348. Doi: 10.4028/www.scientific.net/MSF.503-504.341.
18. Fujioka T, Horita Z (2009) Development of high-pressure sliding process for microstructural refinement of rectangular metallic sheets. *Mater Trans* 50:930–933. Doi: 10.2320/matertrans.MRP2008445.
19. Shin DH, Park JJ, Kim YS, et al. (2002) Constrained groove pressing and its application to grain refinement of aluminum. *Mater Sci Eng A* 328:98-103. Doi: 10.1016/S0921-5093(01)01665-3.
20. Segal VM (2018) Review: modes and processes of severe plastic deformation (SPD). *Materials* 11:1175. Doi: 10.3390/ma11071175.
21. Estrin Y, Vinogradov A (2013) Extreme grain refinement by severe plastic deformation: a wealth of challenging science. *Acta Mater* 61:782–817. Doi: 10.1016/j.actamat.2012.10.038.
22. Valiev RZ, Islamgaliev RK, Alexandrov IV (2000) Bulk nanostructured materials from severe plastic deformation. *Prog Mater Sci* 45:103–189. Doi: 10.1016/S0079-6425(99)00007-9.
23. Beloshenko V, Vozniak I, Beygelzimer Y, et al. (2019) Severe plastic deformation of polymers. *Mater Trans* 60:1192-1202. Doi: 10.2320/matertrans.MF201912.
24. Edalati K (2019) Review on recent advancement in severe plastic deformation of oxides by high-pressure torsion (HPT). *Adv Eng Mater* 21:1800272. Doi: 10.1002/adem.201800272.
25. Révész Á, Kovács Z (2019) Severe plastic deformation of amorphous alloys. *Mater Trans* 0:1283–1293. Doi: 10.2320/matertrans.MF201917.
26. Bridgman PW (1952) *Studies in large plastic flow and fracture*. New York (NY): McGraw-Hill; Doi: 10.4159/harvard.9780674731349.
27. Ivanisenko Y, Kulagin R, Fedorov V, et al. (2016) High pressure torsion extrusion as a new severe plastic deformation process. *Mater Sci Eng A* 664:247–256. Doi: 10.1016/j.msea.2016.04.008.
28. Edalati K, Horita Z (2016) A review on high-pressure torsion (HPT) from 1935 to 1988. *Mater Sci Eng A* 652:325–352. Doi: 10.1016/j.msea.2015.11.074.
29. Zhilyaev AP, Langdon TG (2008) Using high-pressure torsion for metal processing: fundamentals and applications. *Prog Mater Sci* 53:893–979. Doi: 10.1016/j.pmatsci.2008.03.002.
30. Cao Y, Wang YB, Figueiredo RB, et al. (2011) Three-dimensional shear-strain patterns induced by high-pressure torsion and their impact on hardness evolution. *Acta Mater* 59:3903–3914. Doi: 10.1016/j.actamat.2011.03.015.
31. Edalati K, Uehiro R, Fujiwara K, et al. (2017) Ultra-severe plastic deformation: evolution of microstructure, phase transformation and hardness in immiscible magnesium based systems. *Mater Sci Eng A* 701:158–166. Doi: 10.1016/j.msea.2017.06.076.
32. Jiang W, Zhou H, Cao Y, et al. (2020) On the heterogeneity of local shear strain induced by high-pressure torsion. *Adv Eng Mater* 22:1900477. Doi: 10.1002/adem.201900477
33. Blank VD, Popov MY, Kulnitskiy BA (2019) The effect of severe plastic deformations on phase transitions and structure of solids. *Mater Trans* 60:1500–1505. Doi: 10.2320/matertrans.MF201942

34. Bridgman PW (1936) Shearing phenomena at high pressure of possible importance for geology. *J Geol* 44:653-669. <https://www.jstor.org/stable/30056088>.
35. Bridgman PW (1953) The effect of pressure on several properties of the alloys of bismuth-tin and of bismuth cadmium. *Proc Am Acad Arts Sci* 82:101–156. Doi: 10.2307/20023704.
36. Mazilkin AA, Straumal BB, Rabkin E, Baretzky B, Enders S, Protasova SG, Kogtenkova OA, Valiev RZ (2006) Softening of nanostructured Al-Zn and Al-Mg alloys after severe plastic deformation. *Acta Mater* 54:3933-3939. Doi: 10.1016/j.actamat.2006.04.025
37. Zhu YT, Lowe TC, Langdon TG (2004) Performance and applications of nanostructured materials produced by severe plastic deformation. *Scr Mater* 51:825–830. Doi: 10.1016/j.scriptamat.2004.05.006.
38. Zhilyaev AP, McNelley TR, Langdon TG (2007) Evolution of microstructure and microtexture in fcc metals during high-pressure torsion. *J Mater Sci* 42:1517–1528. Doi 10.1007/s10853-006-0628-0.
39. Drai A, Aour B (2013) Analysis of Plastic Deformation Behavior of HDPE during High Pressure Torsion Process. *Eng Struct* 46:87-93. Doi: 10.1016/j.engstruct.2012.06.033.
40. Edalati K, Bachmaier A, Beloshenko VA, et al. (2022) Nanomaterials by severe plastic deformation: review of historical developments and recent advances, *Mater Res Lett* 10:4, 163-256. Doi:10.1080/21663831.2022.2029779.
41. Zhorin VA, Kissin YV, Luiz YV, et al. (1976) Structural changes in polyolefins due to the combination of high pressure and shear deformation. *Polym Sci USSR* 18:3057–3061. Doi: 10.1016/0032-3950(76)90416-0.
42. Goldman AY, Demenchuk NP (1981) Effect of hydrostatic pressure on mechanical properties of crystalline polymers under shear. *Strength Mater* 13:110-114.
43. Beloshenko VA, Voznyak AV, Voznyak YV (2014) Control of the mechanical and thermal properties of semicrystalline polymers via a new processing route of the equal channel multiple angular extrusion. *Polym Eng Sci* 54:531–539. Doi: 10.1002/pen.23583.
44. Beloshenko VA, Voznyak AV., Voznyak YV (2015) Effects of equal-channel, multiple-angular extrusion on the physical and mechanical properties of glassy polymers. *J Appl Polym Sci* 132:42180–42187. Doi: 10.1002/app.42180.
45. Beloshenko VA, Varyukhin VN, Voznyak AV, et al. (2009) New methods of solid-phase modification of polymers by simple-shear deformation. *Dokl Phys Chem* 426:81–83. Doi: 10.1134/S0012501609050017.
46. Vozniak I, Beloshenko V, Savchenko B, et al. (2021) Improvement of mechanical properties of polylactide by equal channel multiple angular extrusion. *J Appl Polym Sci* 138:49720. Doi: 10.1002/app.49720.
47. Zhorin VA, Kulakov VV, Nikolskii VG, et al. (1982) Homogenization of blends of polypropylene and ethylenepropylene rubber under shear in the presence of inorganic fillers. *Polym Sci USSR* 24:1081-1084. Doi: 10.1016/0032-3950(82)90208-8.
48. Zharov AA (2004) Reaction ability of monomers and other organic compounds in solid state under high pressure and shear deformation. *Vysokomoleculyarnya Soedineniya [Macromolecular Compounds]* 46:1613-1637.
49. Figueiredo RB, Aguilar MTP, Cetlin PR, Langdon TG (2012) Analysis of plastic flow during high-pressure torsion, *J Mater Sci* 47:7807-7814. Doi: 10.1007/s10853-012-6506-z.
50. Lee DJ, Yoon EY, Park LJ, Kim HS (2012) The dead metal zone in high-pressure torsion, *Scr Mater* 67:384-387. doi: 10.1016/j.scriptamat.2012.05.024.
51. Bridgman PW (1938) Polymorphic transitions up to 50,000 Kg/cm² of several organic substances. *Am Acad Arts Sci* 72:227-268. Doi: 10.1063/1.1746503.
52. Bridgman PW (1947) The effect of high mechanical stress on certain solid explosives. *J Chem Phys* 15:311-313. Doi: 10.4159/harvard.9780674287839.c38.
53. Edalati K, Bachmaier A, Beloshenoc VA et al. (2022) Nanomaterials by severe plastic deformation: review of historical developments and recent Advances *Mater Res Lett* 10 (4), 163–256. Doi: 10.1080/21663831.2022.2029779

54. Razavi-Khosroshahi H, Edalati K, Arita M, et al. (2016) Plastic strain and grain size effect on high-pressure phase transformations in nanostructured TiO₂ ceramics. *Scr Mater* 124:59-62. Doi: 10.1016/j.scriptamat.2016.06.022.
55. Edalati K, Horita Z (2010) Application of high-pressure torsion for consolidation of ceramic powders. *Scr Mater* 63:174-177. Doi: 10.1016/j.scriptamat.2010.03.048.
56. Zhorin VA, Maksimychyev AV, Ponomarenko AT, et al. (1979) Electrophysical characteristics of several organic compounds at high pressures and high pressures combined with shearing strains. *Bull Acad Sci USSR Div Chem Sci* 28:2078-2082. Doi: 10.1007/BF00947557.
57. Figueiredo RB, Pereira PHR, Aguilar MTP, et al. (2012) Using finite element modeling to examine the temperature distribution in quasi-constrained high-pressure torsion. *Acta Mater* 60:3190-3198. Doi: 10.1016/j.actamat.2012.02.027.
58. Pereira PHR, Figueiredo RB (2019) Finite element modeling of high-pressure torsion: an overview. *Mater Trans* 60:1139-1150. Doi: 10.2320/matertrans.MF201906.
59. Edalati K, Lee DJ, Nagaoka T, et al. (2016) Real hydrostatic pressure in high-pressure torsion measured by bismuth phase transformations and FEM simulations. *Mater Trans* 57:533-538. Doi: 10.2320/matertrans.M2015374.
60. Sue HJ, Dilan H, Li CKY (1999) Simple shear plastic deformation behavior of polycarbonate plate due to the equal channel angular extrusion process. I: finite element methods modeling. *Polym Eng Sci* 39:2505-2515. Doi: 10.1002/pen.11638.
61. Kang F, Wang JT, Su YL, et al. (2007) Finite element analysis of the effect of back pressure during equal channel angular pressing. *J Mater Sci* 42:1491-1500. Doi: 10.1007/s10853-006-0762-8.
62. Kim HS (2001) Finite element analysis of high-pressure torsion processing. *J Mater Process Technol* 113:617-621. Doi: 10.1016/S0924-0136(01)00709-9.
63. Figueiredo RB, Cetlin PR, Langdon TG (2011) Using finite element modeling to examine the flow processes in quasiconstrained high-pressure torsion. *Mater Sci Eng A* 528:8198-8204. Doi: 10.1016/j.msea.2011.07.040.
64. Yoon SC, Horita Z, Kim HS (2008) Finite Element Analysis of Plastic Deformation Behaviour during High Pressure Torsion Processing. *J Mater Process Technol* 201:32-36. Doi: 10.1016/j.jmatprotec.2007.11.204.
65. Kim HS (2001) Finite Element Analysis of High Pressure Torsion Processing. *J Mater Process Technol* 113:617-632. Doi: 10.1016/S0924-0136(01)00709-9.
66. Zhilyaev AP, Nurislamova GV, Kim BK, Baró MD, Szpunar JA, Langdon TG (2003) Experimental parameters influencing grain refinement and microstructural evolution during high-pressure torsion. *Acta Mater* 51(3):753-765. Doi: 10.1016/S1359-6454(02)00466-4.
67. Jiang H, Zhu YT, Butt DP, Alexandrov IV, Lowe TC (2000) Microstructural evolution, microhardness and thermal stability of HPT-processed Cu. *Mater Sci Eng A* 290(1-2):128-138. Doi: 10.1016/S0921-5093(00)00919-9.
68. Jiang H, Zhu YT, Butt DP, Alexandrov IV, Lowe TC (2000) Microstructural evolution, microhardness and thermal stability of HPT-processed Cu. *Mater Sci Eng A* 290:128-138. Doi: 10.1016/S0921-5093(00)00919-9.
69. Vorhauer A, Pippan R (2004) On the homogeneity of deformation by high pressure torsion. *Scripta Materialia*. 51:921-925. Doi: 10.1016/j.scriptamat.2004.04.025.
70. Yang Z, Welzel U (2005) Microstructure-microhardness relation of nanostructured Ni produced by high-pressure torsion. *Mater Lett* 59:3406-3409. Doi: 10.1016/j.matlet.2005.05.077.
71. Chaboche JL (1995) Formalisme général des lois de comportement: Application aux métaux et polymères, dans *Introduction à la mécanique des polymères*. Edition C. G'Sell et J.M. Haudin, Institut national polytechnique de Lorraine. 97-115.
72. Chaboche JL (1997) Thermodynamic formulation of constitutive equations and application to the visco-plasticity and viscoelasticity of metals and polymers. *Int J Solids Struct* 34:2239-2254. Doi: 10.1016/S0020-7683(96)00162-X.